

KOLEJ MATRIKULASI PULAU PINANG

PRE PSPM DP014 SET B 2024/2025

Instruction:

***This question paper consists of 7
subjective questions.***

Attempt all questions .

QUESTION	MARKS
1	/ 8
2	/ 12
3	/ 9
4	/ 12
5	/ 9
6	/ 12
7	/ 18
TOTAL	/ 80

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $v = u + at$ | 16. $a_c = \frac{v^2}{r} = r\omega^2$ |
| 2. $s = ut + \frac{1}{2}at^2$ | 17. $\omega = \omega_0 + \alpha t$ |
| 3. $v^2 = u^2 + 2as$ | 18. $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 19. $\omega^2 = \omega_0^2 + 2\alpha\theta$ |
| 5. $p = mv$ | 20. $\theta = \frac{1}{2}(\omega_0 + \omega)t$ |
| 6. $J = F\Delta t$ | 21. $\tau = rF\sin\theta$ |
| 7. $J = \Delta p = mv - mu$ | 22. $\frac{dQ}{dt} = -kA\left(\frac{dT}{dx}\right)$ |
| 8. $f = \mu N$ | 23. $v_{rms} = \sqrt{\frac{3RT}{M}}$ |
| 9. $W = Fs \cos \theta$ | 24. $v_{rms} = \sqrt{\frac{3kT}{m}}$ |
| 10. $K = \frac{1}{2}mv^2$ | 25. $Q = \Delta U + W$ |
| 11. $U = mgh$ | |
| 12. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$ | |
| 13. $s = r\theta$ | |
| 14. $v = r\omega$ | |

15. $a_t = r\alpha$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Rydberg constant <i>Pemalar Rydberg</i>	R_H	$= 1.097 \times 10^7 \text{ m}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant <i>Pemalar graviti</i>	G	$= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration <i>Pecutan jatuh beba</i>	g	$= 9.81 \text{ m s}^{-2}$

1. (a) A metal block with dimension of $6\text{ cm} \times 12\text{ cm} \times 22\text{ cm}$ and mass of 8.50 kg .

i. What is the volume of the block in m^3 ?

ii. Calculate the density of the metal in kg m^{-3} ?

[2 marks]

(b) **FIGURE 1** below shows forces F_1 and F_2 acting on a particle O.

Calculate the magnitude and direction of the resultant force on particle O.

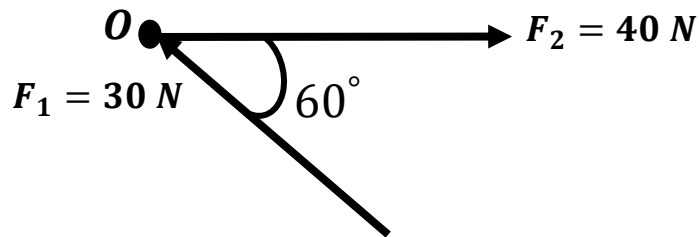


FIGURE 1

[6 marks]

2 (a)

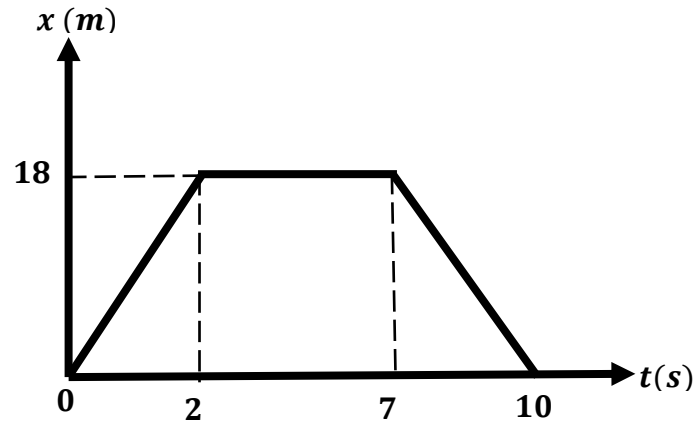


FIGURE 2

FIGURE 2 shows a graph of displacement, x against time, t for a car travelling along a straight road.

(i) Calculate the velocity of the car for the time 0-2 s, 2-7 s and 7-10 s.

(ii) Sketch a velocity against time graph for the whole journey of the car.

[7 marks]

(b) An object moves in a straight line with uniform acceleration of 2 m s^{-2} over a distance of 30 m in 3 s. Calculate the initial velocity of the object.

[2 marks]

(c) A ball is thrown straight upward with an initial velocity of 15 m s^{-1} .

Determine the magnitude and the direction of the ball's displacement after 2 s it is thrown.

[3 marks]

3. a) An egg with a mass of 40 g falls on a ground at speed of 30.67 ms^{-1} .

Later, the egg shatters with time of impact of 0.03 s. Determine

i) the change of momentum experienced by the egg.

ii) the average force exerted on the egg.

[3 marks]

b) A car with mass of 1500 kg is travelling at speed of 75 ms^{-1} when it collides with a Toyota pick-up truck with mass of 2000 kg travelling in opposite direction.

After the collision, both vehicles come to a stop. Determine the velocity of the Toyota pick-up truck just before collision?

[3 marks]

c) A 500 kg lorry moving to the right at speed of 100 m s^{-1} strikes a stationary motorbike of mass 200 kg. After the collision, the lorry rebounds with a speed of 20 m s^{-1} at an angle of 30° above the +ve x axis. Determine the final speed of the motorbike if the motorbike was deflected 6.9° below the +ve x axis.

[3 marks]

4a.

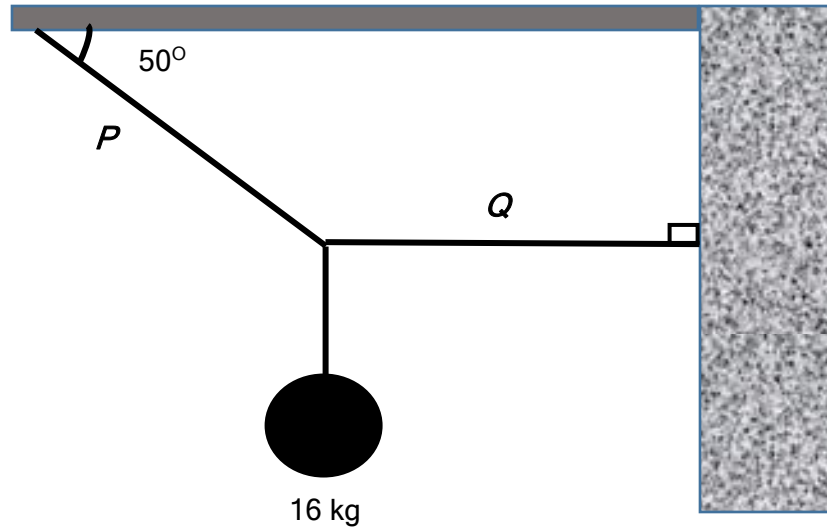


FIGURE 4

FIGURE 4 shows a 16 kg load held in equilibrium by ropes, P and Q fastened to the ceiling and the wall respectively. Calculate the tension P and Q . [4 marks]

b.

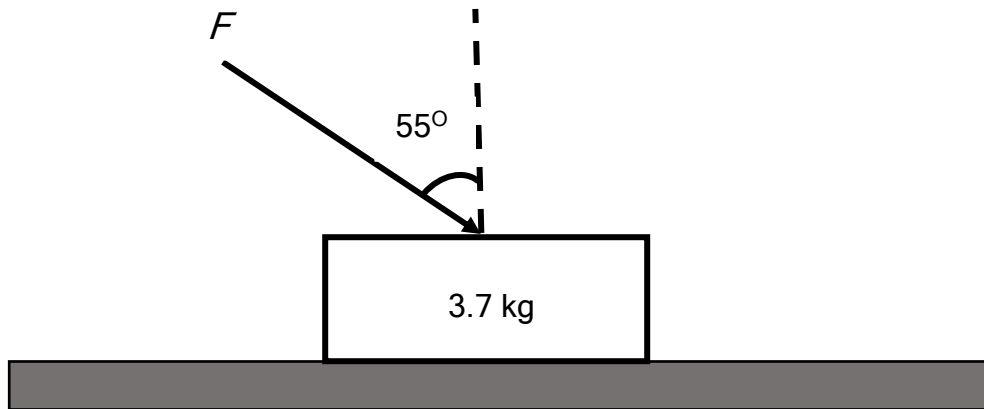


FIGURE 5

FIGURE 5 shows a 3.7 kg block is being pushed along a rough horizontal surface by a force $F = 64 \text{ N}$ at an angle 55° from the normal. The block moves at a constant acceleration 0.8 ms^{-2} .

- i. Sketch a free body diagram for the block.
- ii. Calculate the frictional force.
- iii. Determine the coefficient of friction between the block and the rough surface.

[8 marks]

5 (a) The period of an object that spun around in circular motion is 0.25 s. If the radius is 0.3 m, calculate the

- (i) time take to complete 75 revolutions.
- (ii) angular displacement in 2 minutes.
- (iii) speed of the object.

[4 marks]

(b) A coin is placed on a rotating record player. The speed of the coin is 3.5 m s^{-1} and observed to circulate at rate of 22.4 revolutions per minute. Calculate the

- (i) angular velocity in SI unit.
- (ii) angular displacement of coin in 150 s.
- (iii) the coefficient of friction.
- (iv) radius of the coin is placed.

[5 marks]

6(a). A car is travelling with a velocity of 17.0 ms^{-1} on a straight horizontal highway.

The wheels of the car has a radius of 48.0 cm . If the car then speeds up with an acceleration of 2.00 ms^{-2} for 5.00 s , calculate;

i) the number of revolutions of the wheels during this period,

ii) the angular speed of the wheels after 5.00 s . [5 marks]

(b). If the car that moves with the same initial speed decelerating at the same rate, determine the total number of revolutions required by the wheels to stop.

[3 marks]

(c)

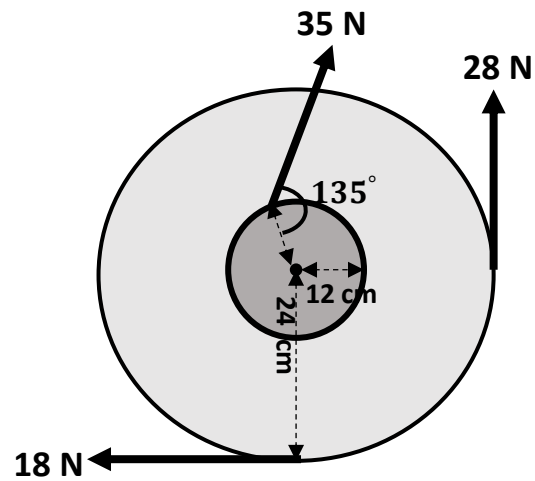


FIGURE 6

A system of 2 concentric wheels is being acted by 3 different forces simultaneously as shown by the figure above. Each force exerts torque on the system.

Find the net torque on the system and state the final direction of rotation of the system. (Neglecting frictions of the system)

[4 marks]

7(a) A 10 cm copper rod has a cross-sectional area 2 cm^2 . One end of the rod is at 100°C and the other end at 30°C . If thermal conductivity of copper is $397 \text{ W m}^{-1} ^\circ\text{C}^{-1}$, determine the

- (i) temperature gradient along the rod.
- (ii) rate of heat flow through the rod.
- (iii) total of heat conducted in 2 minutes.

[6 marks]

(b) A 12 liter tank is filled with compressed oxygen at temperature 25°C .

The molecular mass of oxygen is 32 g mol^{-1} . Calculate the

- (i) rms speed of oxygen molecules.
- (ii) temperature of the oxygen if the rms speed increased to 600 ms^{-1} .

[4 marks]

(c) A gas expands at constant pressure. During the process, 350 J of heat is absorbed and internal energy increases by 150 J .

- (i) Calculate its work done
- (ii) During this process, is the volume increase or decrease?

[3 marks]

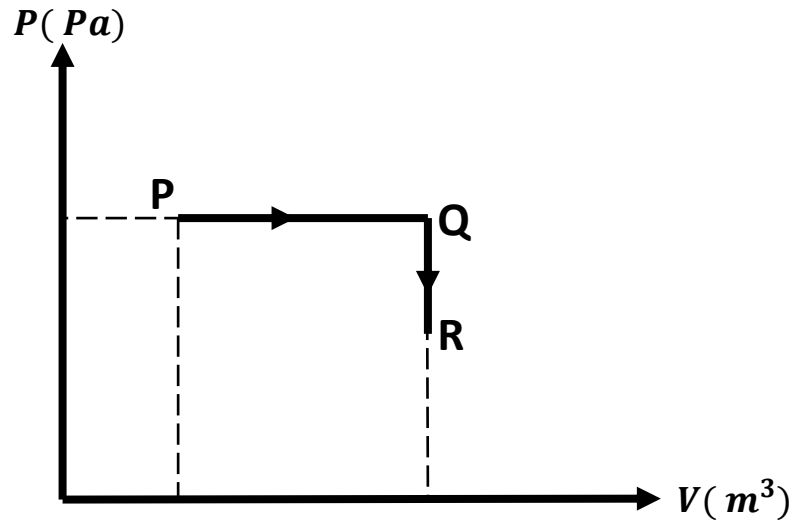


FIGURE 7

(d) An ideal gas undergoes two processes PQ and QR as shown in **FIGURE 7**.

12 kJ of heat is supplied and 6 kJ of work is done when the gas expands during the process from P to Q. A 4 kJ of internal energy decreases during QR process.

- (i) name the process PQ and QR
- (ii) change in the internal energy for PQ.
- (iii) amount of heat absorbed or released for QR process.

[5 marks]

SOALAN TAMAT

